Charged-Sector Defects and Annular Error Certificates for the Critical Planar XY Exponent

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Abstract

The supplied manuscript claims the critical exponent 1/4 for free-box planar XY correlations. This report retains its distinction between ordinary Gaussian height limits, charged Fourier sectors, magnetic partition ratios, and critical endpoint identification. A quantitative refinement of the source's replica and soft-joining argument bounds each charged coefficient by a finite-scale variance deficit and a fourth-cumulant defect. It includes a rational exclusion threshold and explicitly displays the loss from a small phase denominator. A second group of certificates tracks admissible annular test errors, actual magnetic-sector discrepancies, and unequal errors over logarithmically many scales. Weak boundary forms and source-preserving gauge coordinates keep these estimates in physical energy norms. The results do not supply the source's uniform mixed-boundary limits, analytic entry theorem, or critical identification from a finite computation. Contribution assessment: C3/5, the finite charged-sector exclusion bound and explicit annular error budgets are notable source-specific quantitative refinements, conditional on the stated comparison and sector inputs and without an independent global critical-exponent proof.

1 Model, conventions, and the global boundary

Let $\Lambda_L = [-L, L]^2 \cap \mathbb{Z}^2$. Each unoriented nearest-neighbor edge is counted once, and the probability law is

$$Z_{L,b}^{-1} \exp\left\{ b \sum_{\{x,y\}} \cos(\theta_x - \theta_y) \right\} \prod_{x \in \Lambda_L} \frac{\mathrm{d}\theta_x}{2\pi}, \qquad b > 0.$$

There are no exterior edges or prescribed boundary spins. Define

$$C_b(n) = \lim_{L\to\infty} \mathbb{E} \cos(\theta_0 - \theta_{ne_1}), \quad m(b) = \lim_{n\to\infty} -n^{-1} \log C_b(n), \quad b_c = \inf\{b > 0 : m(b) = 0\}.$$

The volume limit is first, at each fixed integer n. The source [1] claims $\log C_{b_c}(n)/\log n \to -1/4$. This is an attributed global claim, not a conclusion established independently here.

The dual height takes values in $2\pi\mathbb{Z}$. A primary edge difference $2\pi j$ has weight

$$p_b(j) = e^{-b} I_j(b), \qquad \sum_{j\in\mathbb{Z}} p_b(j) e^{tj} = \exp\{b(\cosh t - 1)\}.$$

The primary unit-conductance Laplacian A satisfies $\langle g, Ag\rangle = \sum_{\{x,y\}}(g_x - g_y)^2$, with counting-measure pairings. Its continuum form is $\int |\nabla g|^2$, not twice that integral. The effective coefficient a is defined from large-box height variances; it is not the bare parameter b. Gaussian covariance aA^+ corresponds to energy $\langle h, Ah\rangle/(2a)$.

A free height component is modulo translations by $2\pi\mathbb{Z}$. Thus a phase $\exp(i\langle h, s\rangle)$ is representative-independent when $\sum s \in \mathbb{Z}$; a real linear test must have total zero. When confining averages control a constant, its translations are summed instead of choosing a mean-zero representative. In the original free spin box the exterior *dual face* is pinned to zero. This does not fix any primal spin.

The global chain distinguishes four numerical facts:

$$\begin{aligned} \mathcal{E}(\text{radial unit flow}) &= \frac{\log G}{2\pi} + O(1), \qquad a > 0 \Longrightarrow a \geq 8\pi, \\ \mathcal{E}(\text{opposite angular pair}) &= 4\pi \log n + O(1), \quad \eta_{\text{spin}} = \frac{2\pi}{a}. \end{aligned} \tag{1}$$

The gap in a uses a sum of order G^2 positive charge sectors; the magnetic energy alone does not give this gap. Identifying $a(b_c) = 8\pi$ also requires the source's openness above 8π and a nonsharp critical polynomial lower bound.

2 A finite charged-sector certificate

Fix a free m-square. Let u be uniform probability on its vertices and ρ_R, ρ_L its right and left face probabilities. Use the real, representative-independent observations

$$X = \langle h, \rho_R - \rho_L\rangle, \qquad Y = \langle h, \rho_R - u\rangle.$$

Here $X = \langle h, A(x_1/m)\rangle$ exactly, with the source's square convention. Put

$$V = \mathbb{E}X^2, \quad W = \mathbb{E}Y^2, \quad \kappa = \mathbb{E}X^2Y^2 - VW - 2(\mathbb{E}XY)^2.$$

For a nonzero integer l, define

$$F_l(t, v) = \mathbb{E}e^{il\langle h,u\rangle + itX + ivY}, \qquad \alpha_l = F_l(0, 0). \tag{2}$$

The expectation is well-defined on height translation classes. Reflection in the vertical midline changes X to $-X$, fixes $\langle h, u\rangle$ modulo 2π, and implies

$$(F_l)_t(0, 0) = 0, \qquad \mathbb{E}XY = V/2. \tag{3}$$

For the latter equality reflection sends (X, Y) to $(-X, Y - X)$, so $\mathbb{E}XY = -\mathbb{E}XY + V$. This argument uses only neutral observations and does not choose a representative of the height class.

We use the following *source premises*, proved there by finite Gaussian-chain comparison and a fixed-primary-graph limit. Every invariant centered characteristic coefficient is strictly positive and at most one; its logarithmic Hessian in any real mean-free direction is at least minus the centered covariance. The primary observations have the moments needed below. The replica parity identities hold first on the enlarged quadratic lattice and their resulting inequalities pass to the Bessel law. These premises are not true for arbitrary moment data.

Lemma 2.1 (Quantitative replica identity)**.** *Under these premises, $\kappa \geq 0$, and for all real t, v,*

$$|(\log F_l)_{tv}(t, v) + V/2| \leq \frac{\sqrt{\kappa/2}}{F_l(t, v)^2}. \tag{4}$$

Proof. On a finite quadratic lattice take independent h_1, h_2 and put $S = h_1 + h_2, D = h_1 - h_2$. Given their common parity class modulo 4π, they are conditionally independent and identically distributed. For the observations X, Y let $Z = \mathbb{E}(S_X S_Y \mid \text{parity})$. Direct expansion gives

$$\mathbb{E}Z = 2\mathbb{E}XY, \qquad \operatorname{Var} Z = 2\{\mathbb{E}X^2Y^2 - \mathbb{E}X^2\mathbb{E}Y^2 - 2(\mathbb{E}XY)^2\} = 2\kappa.$$

For example, $\mathbb{E}S_X S_Y D_X D_Y = 2\mathbb{E}X^2Y^2 - 2VW$; subtracting $4(\mathbb{E}XY)^2$ gives the displayed variance. Conditional independence and differentiating the two-copy transform give

$$-2\{F_l(F_l)_{tv} - (F_l)_t(F_l)_v\} = \mathbb{E}[Z\,\mathbb{E}(e^{i\theta(D)} \mid \text{parity})], \quad \theta(h) = l\langle h, u\rangle + tX + vY.$$

The conditional phase has modulus at most one, and its expectation is F_l^2. Replacing Z by $2\mathbb{E}XY$ therefore costs at most $\sqrt{2\kappa}$. Divide by $2F_l^2$ and use (3). The exact finite-graph chain limit passes all fourth moments and transform derivatives; strict positivity allows the division on each fixed compact parameter set. □

For precision, specify the actual two-cell soft join. It sums the relative lattice translations with the scalar Gaussian mismatch penalty used in the source. Its normalized transform for the sum of the two rightward internal affine observations has the form

$$\Phi(t) = \frac{\sum_{j\in\mathbb{Z}} e^{-j^2/2}\chi_j F_j(t,j)^2}{\sum_{j\in\mathbb{Z}} e^{-j^2/2}\chi_j F_j(0,j)^2}, \qquad 0 < \chi_j \le 1, \quad \chi_{-j} = \chi_j, \quad \chi_0 = 1. \tag{5}$$

The χ_j are the actual independent-seam characteristic factors. Reflection of the second cell followed by height inversion explains the square, with the same sign of t. In particular Φ is the characteristic function of a centered real test, not an arbitrary positive sector sum. Write

$$V_{\text{soft}} = -\Phi''(0), \qquad \Delta = 2V - V_{\text{soft}} \ge 0, \qquad \Theta = \sum_{j\in\mathbb{Z}} e^{-j^2/2}. \tag{6}$$

The exact join, its comparison inequalities, and the seam law are essential inputs. A numerical replacement of the sector weights must first be given rigorous error bounds.

Theorem 2.2 (Finite exclusion of a charged coefficient)**.** *Suppose $V > 0$ and the preceding source premises hold. Fix $l \in \mathbb{Z} \setminus \{0\}$ and $0 < \epsilon \le 1$. Set*

$$q_\epsilon = \exp\left[-\left(\sqrt{\log(1/\epsilon)} + |l|\sqrt{W/2}\right)^2\right], \qquad B_l = \frac{2e^{-l^2/2}\chi_l}{\Theta}. \tag{7}$$

If $\alpha_l \ge \epsilon$, then

$$\Delta \ge B_l l^2 q_\epsilon^2 \left(V - \frac{\sqrt{2\kappa}}{q_\epsilon^2}\right)_+^2. \tag{8}$$

Consequently any strict violation of this inequality certifies $\alpha_l < \epsilon$. Certified lower bounds for B_l, q_ϵ, V and an upper bound for κ may be substituted.

Proof. The logarithmic-Hessian comparison in the Y direction gives the source's phase-transfer inequality

$$\sqrt{-\log F_l(0,v)} \le \sqrt{-\log \alpha_l} + |v|\sqrt{W/2}.$$

One proof is to put $f = \log F_l \le 0$, use $f'' \ge -W$ to obtain $|f'|^2 \le -2Wf$, and integrate; a positive regularization handles zeros of $-f$. Thus $F_l(0,v) \ge q_\epsilon$ on the interval joining 0 to l. Integrating (4) and using $(F_l)_t(0,0) = 0$ gives the sector slope

$$d_l = \frac{\mathrm{d}}{\mathrm{d}t} \log F_l(t,l)^2 \Big|_{t=0}, \qquad |d_l| \ge |l| \left(V - \frac{\sqrt{2\kappa}}{q_\epsilon^2} \right)_+ .$$

At $t = 0$ let p_j be the normalized positive weights in (5). The logarithmic second derivative of each square is at least $-2V$. The logarithmic derivative of the whole sum is zero, by inversion symmetry. The mixture-Hessian identity yields

$$\Delta \ge \sum_j p_j d_j^2.$$

These differentiations are justified, not formal: raw second derivatives are bounded by $2V$, and if $0 < \psi \le 1$ and $(\log \psi)'' \ge -2V$, then $\psi |(\log \psi)'|^2 \le 4V(-\psi \log \psi) \le 4V/e$. The Gaussian sector weights dominate all the resulting sums. The denominator in (5) is at most Θ. The two sectors $l, -l$ have equal weights and opposite slopes. Together they give (8). For substitution of bounds, note that $(Vq - d/q)_+^2$ increases with $V, q > 0$ and decreases with $d \ge 0$. □

2.1 An explicit bound and a rational exclusion example

Corollary 2.3. *With the hypotheses of Theorem 2.2, set*

$$d = \sqrt{2\kappa}, \quad u_l = \frac{\sqrt{\Delta / B_l}}{|l|}, \quad Q_l = \frac{u_l + \sqrt{u_l^2 + 4Vd}}{2V}.$$

Then

$$\alpha_l \le \min \left\{ 1,\ e^{l^2 W/2} \sqrt{Q_l} \right\}. \tag{9}$$

If V is bounded below, W bounded above, and B_l bounded below for a fixed l, this is $\alpha_l = O_l(\Delta^{1/4} + \kappa^{1/8})$. No decay rate in the primary side length is assumed.

Proof. Use $\epsilon = \alpha_l$. Since $(x+y)^2 \le 2x^2 + 2y^2$, the phase bound is at least $q = \alpha_l^2 e^{-l^2 W}$. The theorem and monotonicity imply $(Vq - d/q)_+ \le u_l$. Hence $Vq^2 - u_l q - d \le 0$, so $q \le Q_l$. The power estimate follows by bounding the positive root. If $d = \Delta = 0$, the conclusion is obtained by the same inequality: no positive α_l is possible. □

For an exact envelope example, suppose the certified data satisfy

$$l = 1, \quad V \ge 1, \quad W \le \tfrac{1}{2}, \quad \chi_1 \ge \tfrac{1}{2}, \quad \kappa \le \tfrac{1}{524288}, \quad \Delta \le \tfrac{1}{1024}.$$

The necessary covariance condition is $W \ge V/4$, by $\mathrm{Cov}(X,Y) = V/2$ and Cauchy–Schwarz. This envelope permits, for example, $V = 1, W = 1/2$; it does not prescribe an inconsistent small value of W. If $\alpha_1 \ge 1/2$, the relaxed phase lower bound is

$$q \ge \tfrac{1}{4} e^{-1/2} \ge \tfrac{1}{8}, \qquad d \le \tfrac{1}{512}, \qquad B_1 \ge \tfrac{1}{6}.$$

For the last inequality, $e^{-1/2} \ge 1/2$ and $\Theta < 3$. To check the latter without a numerical theta oracle, the first positive term is at most $2/3$, and successive terms have ratio at most $e^{-3/2} < 1/4$;

use the first four terms of the positive Taylor series for $e^{3/2}$ to verify that last bound. Thus $\Theta \le 1 + 2(2/3)/(1 - 1/4) = 25/9 < 3$. Theorem 2.2 would force

$$\Delta \ge \frac{1}{6}\frac{1}{64}\left(1 - \frac{1}{8}\right)^2 = \frac{49}{24576} > \frac{1}{1024}, \tag{10}$$

a contradiction. This is a conditional numerical certificate, not a claim that a particular finite XY box has these data. All moments, the seam factor, and the soft-join variance would need certified enclosures for such an application.

2.2 From charged bounds to phase tests

The certificate concerns every *fixed* nonzero integer l. It is not uniform as $|l| \to \infty$: both the Gaussian sector weight and the phase-transfer denominator deteriorate. The following consequence makes the truncation cost explicit.

Proposition 2.4 (Finite Fourier testing)**.** *Let $M = \langle h, u\rangle \bmod 2\pi$, and suppose $|\mathbb{E}e^{ilM}| \le A_l$ for $1 \le l \le K$. For any real 2π-periodic function f with Lipschitz constant L_f in angular distance,*

$$\left|\mathbb{E}f(M) - \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,\mathrm{d}t\right| \le L_f\left\{\frac{\pi[\,1/2 + \log(K+1)\,]}{K+1} + 2\sum_{l=1}^{K}\left(1 - \frac{l}{K+1}\right)\frac{A_l}{l}\right\}. \tag{11}$$

Proof. Convolve with the Fejer kernel $F_K(t) = (K+1)^{-1}[\sin((K+1)t/2)/\sin(t/2)]^2$, normalized against $\mathrm{d}t/(2\pi)$. For $0 < |t| \le \pi$,

$$F_K(t) \le \min\{K+1,\ \pi^2/[(K+1)t^2]\}.$$

Splitting the first angular moment at $\pi/(K+1)$ gives $\int |t|F_K(t)\,\mathrm{d}t/(2\pi) \le \pi[1/2+\log(K+1)]/(K+1)$. Lipschitz functions are absolutely continuous and $|\widehat{f}(l)| \le L_f/|l|$ by integration by parts. The Fejer polynomial has coefficient $(1 - |l|/(K+1))\widehat{f}(l)$ and the same constant coefficient as f. Bound its expected nonconstant part using the A_l; the single approximation error gives (11). □

This is classical Fourier approximation with new inputs from (8); the Fejer kernel itself is not a new method. For a joint phase with a mean-free real test Z, the source comparison also gives the useful upper transfer

$$\mathbb{E}e^{ilM+iZ} \le \exp\left\{-\left(\sqrt{\log(1/\alpha_l)} - \sqrt{\mathrm{Var}(Z)/2}\right)_+^2\right\}. \tag{12}$$

Apply the phase-transfer inequality in reverse, from $lM + Z$ back to lM. The coefficient is positive by the same source premise. A primary flow certificate $\mathrm{Var}(Z) \le C_b \sum_e w_e^2$ can be used here, but it must include the actual boundary and all source charges.

3 Annular testing with quantified profile losses

Let $\mathcal{A} = \{r < |z| < R\}$, $\tau = \log(R/r) > 0$, and $v = \nabla \arg z = (-y, x)/|z|^2$. Then

$$\|v\|_{L^2(\mathcal{A})}^2 = 2\pi\tau.$$

A radial tangential form $p = w(\log(|z|/r))v$, with w smoothly supported in $(0, \tau)$, is divergence free and has no normal flux. Its stream-function discretization on faces is therefore a legitimate local test in the source's spin-upper argument. Put

$$D = \langle p, v \rangle = 2\pi \int_0^\tau w(t)\,\mathrm{d}t, \qquad E = \|p\|_2^2 = 2\pi \int_0^\tau w(t)^2\,\mathrm{d}t. \tag{13}$$

These are physical integrals, not norms of unweighted polynomial coefficients. The deterministic pairing with a 2π magnetic period is D, after the source's lattice approximation.

Proposition 3.1 (Exact profile deficit)**.** *For any nonzero $p \in L^2(\mathcal{A}; \mathbb{R}^2)$,*

$$\frac{D^2}{E} = 2\pi\tau - \min_{c\in\mathbb{R}} \|v - cp\|_2^2\,. \tag{14}$$

In particular, if $\|p - v\|_2 \le \epsilon\sqrt{2\pi\tau}$ with $0 \le \epsilon < 1$, then

$$\frac{D^2}{E} \ge (1 - \epsilon^2)2\pi\tau, \qquad D > 0.$$

If a computed profile $\widetilde{p}$ and physical error $\|p - \widetilde{p}\|_2 \le e$ are certified, then

$$|D - \widetilde{D}| \le e\sqrt{2\pi\tau}, \qquad \sqrt{E} \le \sqrt{\widetilde{E}} + e.$$

Hence a positive lower bound for D, divided by the latter energy upper bound, gives a certified lower bound for D^2/E.

Proof. Expand the squared norm as $2\pi\tau - 2cD + c^2E$ and minimize. Its minimum is no larger than its value at $c = 1$. Also $D \ge 2\pi\tau - \|p - v\|_2\sqrt{2\pi\tau} > 0$. The final two estimates are Cauchy–Schwarz and the triangle inequality. □

The one-dimensional projection identity is classical; its useful role is that profile error affects the optimized annular exponent quadratically, instead of treating pairing and energy errors as unrelated first-order losses. Admissibility is separate: a small L^2 error does not enforce zero divergence or the correct boundary support.

For a completely explicit profile, leave collars of width $\delta > 0$ in logarithmic coordinates, put $T = \tau - 2\delta$, and use a trapezoid on the remaining interval, with linear ramps of length h, $0 < h \le T/2$, and height one. Direct integration gives

$$\frac{D^2}{2\pi E} = \frac{(T-h)^2}{T - 4h/3}, \qquad \tau - \frac{D^2}{2\pi E} = 2\delta + \frac{(2/3)Th - h^2}{T - 4h/3}. \tag{15}$$

The piecewise affine profile is a limit of smooth profiles supported inside the same annulus; its L^2 approximation error can be charged by Proposition 3.1. It is not silently substituted for a smooth profile in a lattice convergence theorem. With $\delta = \tau/20$, $h = \tau/10$, the captured fraction is exactly $96/115$, and the lost fraction $19/115$. This example works at $\tau = \log 2$ without evaluating that transcendental constant numerically.

3.1 Unequal scales and optimized tilts

The next statement isolates precisely the finite estimate required from the source's cut-and-tilt argument. Let $\mathcal{J}(n)$ index the disjoint annuli around both sources. Suppose actual finite-volume comparisons, uniform in sufficiently distant free outer boundaries, give

$$\log C_b(n) \le \sum_{j\in\mathcal{J}(n)} \{-\lambda_j D_j + \tfrac{1}{2}U_j\lambda_j^2 + \eta_j\} \quad (0 \le \lambda_j \le L_j), \tag{16}$$

where $U_j > 0$, $D_j \ge 0$, $\eta_j \ge 0$ are certified, and all remainders for the allowed tilts have been included. Passing to $C_b(n)$ is justified only by the stated volume-uniform comparison. One cannot deduce (16) from measured unconditioned Gaussian moments.

Theorem 3.2 (Annular tilt budget)**.** *Under* (16)*, set* $\lambda_j^* = \min\{D_j/U_j, L_j\}$*. Then*

$$\log C_b(n) \le -\sum_j \{\lambda_j^* D_j - \tfrac{1}{2}U_j(\lambda_j^*)^2\} + \sum_j \eta_j. \tag{17}$$

If all $D_j/U_j \le L_j$*, the gain is* $\frac{1}{2}\sum_j D_j^2/U_j$*. It is at least the gain from the best single common unrestricted tilt, and the excess is exactly*

$$\frac{1}{2}\left\{\sum_j \frac{D_j^2}{U_j} - \frac{(\sum_j D_j)^2}{\sum_j U_j}\right\} = \frac{1}{2}\sum_j U_j\left(\frac{D_j}{U_j} - \frac{\sum_i D_i}{\sum_i U_i}\right)^2. \tag{18}$$

Proof. Each scalar quadratic in (16) is minimized at the stated clipped value. Expanding the square proves the second identity, including its sign. □

For instance, $U_1 = U_2 = 1, D_1 = 1, D_2 = 2$ gives gain $5/2$ instead of $9/4$. This is a certified-envelope example, not a measured XY improvement. The actual gain must survive any tilt-dependent increase of the η_j.

For a fixed $a > 0$, if $U_j = aE_j$ and the normalized profile losses are ϵ_j^2, the unrestricted gain is at least

$$\frac{\pi}{a}\sum_j (1-\epsilon_j^2)\tau_j.$$

At both sources $\sum_j \tau_j = 2\log n + O(1)$, up to fixed inner and outer exclusions. Thus the exponent loss is bounded by

$$\frac{\pi}{a\log n}\sum_j \epsilon_j^2\tau_j + \frac{\sum_j \eta_j}{\log n} + O(1/\log n). \tag{19}$$

Only logarithmic averages need vanish. A finite number of poor scales with bounded cost is harmless; a positive average error is not. The source proves convergence at each fixed admissible test and then uses Cesaro averaging. Without a uniform approximation theorem, one may not choose increasingly oscillatory profiles or unbounded λ_j as j grows.

4 Primary flows, boundary loads, and gauges

Let D be an oriented incidence matrix taking vertex potentials to edge differences on a connected finite primary graph, so $A = D^T D$. For a real load f of total zero, set $\mathcal{R}(f) = f^T A^+ f$. A flow w is admissible if $D^T w = f$. Dirichlet pins are implemented by restricting the potential space and the incidence columns, not by deleting the energy of edges ending at a pin. With several free components, neutrality and one gauge are required separately on each component.

Proposition 4.1 (A sharp primary variance prefactor)**.** *Assume the source's centered Gaussian-chain comparison for a primary Bessel height law, with any additional centered positive precision or zero constraints. For* $X = \langle f, h\rangle = w^T Dh$*,*

$$\log \mathbb{E}e^{tX} \le b\sum_e \{\cosh(2\pi t w_e) - 1\}, \qquad t \in \mathbb{R}, \qquad \mathrm{Var}(X) \le 4\pi^2 b\, \mathcal{R}(f). \tag{20}$$

For any such flow, write $R_w = \sum w_e^2$ *and* $B_w = \max |w_e|$*. If* $B_w > 0$*, put* $\lambda = bR_w/B_w^2$*,* $\sigma = 2\pi B_w$*. Then for* $x \ge 0$*,*

$$\mathbb{P}(|X| \ge x) \le \min\{1,\ 2e^{-\lambda H(x/(\lambda\sigma))}\}, \quad H(z) = z\,\mathrm{arsinh}\, z - \sqrt{1+z^2} + 1. \tag{21}$$

If $B_w = 0$*,* $X = 0$*. The constant* $4\pi^2 b$ *is sharp within the class of all such finite primary graphs.*

Proof. On each finite Gaussian-chain approximant release endpoint, curl, and period constraints and the added centered precision. Centered transform comparison bounds the transform by that of independent complete-chain increments. Take the subdivision limit at the fixed primary graph. An independent increment $2\pi J$, $J \sim p_b$, has logarithmic transform $b(\cosh(2\pi t) - 1)$, giving the first inequality. Both sides vanish with their first derivative at zero; comparing their second-order expansions gives $\mathrm{Var}(X) \le 4\pi^2 bR_w$. Minimize over admissible flows. For a single free primary edge and its difference test, equality holds, proving sharpness in the stated graph class.

The positive power series for cosh and $|w_e|^{2k} \le B_w^{2k-2} w_e^2$ give

$$b\sum_e (\cosh(2\pi t w_e) - 1) \le \lambda(\cosh(\sigma t) - 1).$$

Optimize the Chernoff bound at $\sigma t = \mathrm{arsinh}(x/(\lambda\sigma))$; apply the same argument to $-X$. No quadratic-exponential moment or uniform microscopic-chain resistance was used. □

This extracts an explicit constant from the source's $C_b\mathcal{R}(f)$ estimate. It uses its finite comparison, not a new proof of that comparison. It does not assert the same upper bound under arbitrary nonzero pin data or real tilts: affine covariance has the opposite lower comparison. Nor can the fourth cumulant κ in Theorem 2.2 be replaced by the fourth cumulant of the released independent increments. Transform domination alone does not give that cumulant comparison.

Proposition 4.2 (Primal and flow enclosure)**.** *For any admissible* w *and any potential* v *in the specified gauge or pinned space,*

$$2f^T v - v^T A v \le \mathcal{R}(f) \le \|w\|_2^2, \qquad \|w\|_2^2 - (2f^T v - v^T A v) = \|w - Dv\|_2^2. \tag{22}$$

Proof. Let u solve $Au = f$ in that space. Then

$$\mathcal{R}(f) - (2f^T v - v^T A v) = \|D(u - v)\|_2^2.$$

Also $w - Du$ is orthogonal to every admissible gradient, since $D^T(w - Du) = 0$. Thus $\|w\|^2 - \mathcal{R}(f) = \|w - Du\|^2$, and the gap identity follows either by Pythagoras or direct expansion. □

These classical Dirichlet–Thomson identities provide supporting certificates for W and the transfer cost in (12). A nonzero divergence residual is not an admissible flow. It must first be repaired with a rigorously bounded additional flow; a small unweighted residual does not suffice.

For a free graph and a real gauge vector μ with $\mathbf{1}^T\mu = 1$, the original Ram derivation [8] gives

$$G_\mu = (I - \mathbf{1}\mu^T)A^+(I - \mu\mathbf{1}^T), \quad f^T G_\mu g = f^T A^+ g \quad (\mathbf{1}^T f = \mathbf{1}^T g = 0). \tag{23}$$

This is gauge independence of *neutral pairings*, not permission to discard a charged phase. A height character of integer total remains well-defined modulo 2π; its nonzero total cannot be treated as a zero-total real test without subtracting an integer point charge in the phase.

For a finite admissible basis with reconstruction matrix R, stiffness and dual loads become $R^T AR$ and $R^T f$. Both must be transformed. Eliminating an interior set I with all loads supported in R_0 preserves the neutral pairing through the classical Schur complement

$$A_{\mathrm{eff}} = A_{R_0R_0} - A_{R_0I}A_{II}^{-1}A_{IR_0}.$$

If interior loads are present, their Schur-transferred loads and interior energy terms must also be retained. These are supporting identities from the freshly read original, not new M396 contributions. In particular, the Gaussian bare energy of many tiny subdivision bonds is not the primary flow norm in (20).

5 Magnetic sectors and logarithmic accumulation

A real lift on a cut punctured domain must have exact periods $+2\pi, -2\pi$ around the two holes. Its branch jumps, the integer height connection, and the observation targets are changed by the same gauge. On a simply connected chart, an integer connection ∇k is removed by $h' = h + k$; a real numerator tilt must use $\nabla h + \nabla k$, so that the same substitution removes the connection from the tilt too. Otherwise a spurious scalar is introduced.

There is an elementary obstruction to ignoring this condition. On an oriented four-edge cycle an exact discrete gradient sums to zero, whereas the connection $(2\pi, 0, 0, 0)$ has sum 2π. No gauge removes it. On a cycle with N edges the constant connection $2\pi/N$ has norm $2\pi/\sqrt{N} \to 0$, but still has period 2π. Therefore a small coefficient or energy residual does not certify the zero sector, uniformly over refining cycles.

Theorem 5.1 (Lower-exponent error ledger)**.** *Fix b with $a(b) = a > 0$. Suppose that for each $n \geq 2$ the actual punctured, observation-weighted magnetic and centered partition functions satisfy, uniformly in sufficiently distant outer boundaries,*

$$C_{L,b}(n) \geq c_n \frac{Z_n^s}{Z_n^0}, \qquad c_n > 0, \quad \left|\log \frac{Z_n^s}{Z_n^0} - \log \frac{Z_{g,n}^s}{Z_{g,n}^0}\right| \leq \sum_{j \in \mathcal{J}_n} \eta_{n,j}.$$

Assume the real Gaussian ratio is the exponential of minus one half its minimum affine energy. Let an admissible trial have energy at most

$$a^{-1}\{4\pi \log n + R_n\} + H_n, \qquad H_n \geq 0,$$

where H_n includes its observation mismatch. Then

$$\frac{\log C_b(n)}{\log n} \geq -\frac{2\pi}{a} - \frac{R_n/(2a) + H_n/2 + \sum_j \eta_{n,j} - \log c_n}{\log n}. \tag{24}$$

It suffices for the numerator of the error fraction to be $o(\log n)$ to retain the source's lower exponent.

Proof. The Gaussian minimum is at most the trial energy. Take logarithms of the three genuine inequalities, add them, pass to the free-box limit at fixed n, and divide by $\log n > 0$. □

The theorem is useful error bookkeeping, not an alternative derivation of the sector comparison. In the source $|\mathcal{J}_n| = O_M(1 + \log n)$, every local discrepancy is at most $\eta_R \to 0$, the microscopic holes are fixed before $n \to \infty$, $H_n = 0$ for the chosen target, and $-\log c_n$ is bounded by a constant depending on those fixed parameters. Each assertion must remain in force.

The weighted formulation permits unequal local accuracies. For example, if there are at most $Aq + B$ events through annular level q, and the total error at level j is at most $K/(j + 1)$, then $\sum_j \eta_{n,j} = O(\log \log n)$ when $q = O(\log n)$. If instead a fraction d of the levels has total error $\eta > 0$, its asymptotic loss is proportional to $d\eta$; the number of levels, not the number of microscopic vertices, is the relevant count. These statements are conditional on the *actual* exact isolation expansion having these bounds. Changing the localization radius at each scale also changes collars, constants, and the fixed-hole threshold; one must not infer admissibility from a summable numerical error schedule alone.

5.1 Weak energy certification with the period fixed

For a fixed admissible lift v_0, variations w belong to the single-valued energy space with the prescribed outer trace and any hard pins. Free walls have the weak Neumann condition. On a component with no trace pin, observation averages must control its constant, or a quotient must be explicitly chosen. Set

$$T(w, z) = a^{-1} \int \nabla w \cdot \nabla z + \langle Bw, Bz \rangle, \quad \ell(z) = -a^{-1} \int \nabla v_0 \cdot \nabla z.$$

The energy relative to target Bv_0 is

$$\mathcal{F}(w) = a^{-1} \|\nabla v_0\|_2^2 - 2\ell(w) + T(w, w).$$

Coercivity on this specified space gives a unique w_* with $T(w_*, z) = \ell(z)$, and exactly

$$\mathcal{F}(\widetilde{w}) - \min \mathcal{F} = \|\widetilde{w} - w_*\|_T^2 = \|\ell - T(\widetilde{w}, \cdot)\|_{T^*}^2 . \tag{25}$$

Expand the quadratic and use the weak equation to prove it. Finite conforming minimization gives an upper energy trial. An upper bound for the *full* dual residual gives the opposite enclosure; finitely many residual tests give only a lower bound for that dual norm. This applies the original weak/coercive Ram treatment [6], with classical Hilbert-space arguments. It cannot change a period, repair missing trace data, or prove that the non-Gaussian ratio is close to the Gaussian one.

6 Source architecture and imported obligations

The whole cached scientific manuscript was read, including its two figure sources and reference locator file. The dependency map below records what supports the use of its premises; it is not a claim that every global argument was independently reproved or that the cited companion was read in full.

Duality and free limits. The main file fixes the ordinary cosine coupling and free-box order. The heights module first obtains exterior-zero duality with opposite integral circulations. Bessel finite comparisons are transferred using long Gaussian chains with parameter $q_N = b/(2N)$, at each *fixed* primary graph. Affine supports and numerator-only real tilts are retained. Whole-chain moments, not the microscopic inverse precision, control seams. Cutting, folding, and saturation then identify the scalar free coefficient and all specified Gaussian smear limits. The source proves fractional-mean uniformity by its replica/soft-joining contradiction; Theorem 2.2 quantifies that particular step.

The coefficient gap. In the source's larger soft join, central single-charge sectors number at least cG^2. A prescribed linear flow right inverse works for all these sectors. The radial unit-flow odd-face cost is $(2\pi)^{-1}\log G + O(1)$, while the even-face cost is bounded. After the primary limit, positivity gives

$$\liminf_{m\to\infty} \frac{z_{Gm}}{z_m} \geq c_b G^{2-a/(4\pi)}, \qquad 0 < z_m \leq 1.$$

For $0 < a < 8\pi$, choose one sufficiently large fixed integer G, obtain an eventual ratio larger than one, and iterate. No observed finite collection of ratios implies this eventual inequality. The coefficient $1/(2\pi)$ in the continuum radial energy and the uniform face discretization remainder remain indispensable.

Mixed pins and typical data. The pins module uses finite face-connected rectangular geometries; point contacts do not identify components. Its continuum space is

$$\{v \in H^1(U) : v = 0 \text{ a.e. on the positive-area pin } p\}, \qquad a^{-1}\int_U |\nabla v|^2.$$

Hard pins constrain primary heights, not all internal cable vertices. The limits are primary mesh first, band-cell mesh second, and band strength last. Cable exploration, polynomial damping, actual weighted moments, and truncation of large tilts give vanishing reach. A relative estimate, not an absolute pin-probability estimate, handles rare hard pins. Conditional laws are approximated through fixed finite probe lists at *typical* guard data. The lists grow after refinement. The Gaussian separated-pin determinant and locality arguments are imported; they are not scored again here. The centered identity $\mathbb{E}r_H = e^{-D}$ is not asserted under tilted sampling.

Magnetic sectors. The magnetic module constructs compatible annular patches for all integer separations, with fixed holes before the separation limit. Exact marked isolation and skip rules give signed patterns. The proof separately verifies centered telescoping, typical-guard gains, dependence factorization, positivity of restricted polymer sums, and local rim insertions. An extensive partition estimate alone would not imply volume-independent rim tightness. Those inputs justify filling the holes at bounded cost and comparing genuine winding-sector ratios. Ordinary smear convergence alone does neither.

Openness and the endpoint. The openness module strengthens *block averages*, not microscopic heights, to a fixed reference $a' = (a^{-1} + \delta)^{-1} > 8\pi$. It proves local pin limits, large-field reserve, and entry into a full analytic activity norm including complex directions and all derivatives. Uniform reference Gaussian curves and a torus variance test then remove the added precision. This is not continuity of the unknown function $a(b)$. The critical module uses the established nonsharp bound $C_{b_c}(n) \geq 1/(8n)$ to rule out $a(b_c) = 0$, the gap to obtain $a(b_c) \geq 8\pi$, and openness plus the magnetic lower exponent to rule out strict inequality. Finally both spin exponents give $1/4$. All these global links are retained as attributed source obligations. Logarithmic corrections to $n^{-1/4}$ are compatible with the claim.

7 Original Ram connections and primary literature

The original weak-operator treatment [6] proves the coercive inverse on a named Sobolev space, derives the sign of Neumann compatibility, and separates a full dual residual from its finite tests.

Here those distinctions specify the admissible space in (25) and the boundary loads in (22). A free-space inverse would not enforce the source's mixed pins and free walls.

The freshly read transform chapter [7] defines a positive-analysis weighted Fourier transform on a specified vertical contour, with inverse factor $(2\pi)^{-d}$ and weighted Plancherel norm. It explicitly retains boundary terms and distinguishes an exact spectral representation from a diagonalization. In this report the circle analysis measure is $\mathrm{d}t/(2\pi)$, the height period is 2π, and only integer total characters survive gauge changes. These conventions explain the Fourier normalization in (11); the continuous transform of a general function is not substituted for a lattice sector sum. The chapter's included child files were not read.

The original general-gauge and elimination proofs [8] justify (23) and source-aware reduction. Only their finite graph statements are transferred; their arithmetic height interpretation is unrelated to the XY model. All three connections use actual original proof passages, not catalogue labels as evidence. Classical coercivity, Fourier analysis, electrical flow duality, and Schur reduction retain their classical attribution.

Primary public records were checked on October 9, 2026. Van Engelenburg–Lis [2], Theorem 1 in the primary text, gives exponential decay below a positive finite threshold and $1/(8|x-y|)$ as a lower bound at and above it, in the matching free-boundary cosine convention. This is not the exponent 1/4. Its theorem statement and model were checked; the full external proof was not independently audited.

Lammers' July 25, 2026 version [3] states a qualitative effective-temperature gap, exponential localization, and delocalization at the threshold. Its abstract does not identify the sharp 8π coefficient used here. Durand–Lammers [4], posted May 27, 2026, relates the XY mass to the right derivative of dual free energy, with a uniform quadratic error near zero slope; that statement does not replace the source's magnetic-sector comparison or its openness proof. These records were used for precise status and scope, not as freshly verified full proofs.

Regev–Stephens-Davidowitz [5] is the classical lattice-Gaussian inequality underlying the source's comparison chain. The public abstract and bibliographic record were checked. The source's specific affine, tilted, and Bessel transfers remain attributed inputs here. A release date or a public preprint title does not by itself certify an endpoint theorem.

8 Advantages, limitations, and next questions

Construction	What becomes quantitative	Remaining requirement
Replica and soft join	Fixed charged coefficient from $\Delta, \kappa, V, W, \chi_l$	Actual Bessel moments and exact soft sector law
Fourier tests	Finite phase tests with a displayed Fejer remainder	Large-charge bounds are not uniform
Annular profile and tilts	Projection loss and unequal-scale gain	Admissible divergence-free tests and uniform transform errors
Primary flows	Exact $4\pi^2 b$ variance prefactor and flow gap	Centered law, correct loads and pins
Magnetic ledger	Separate energy, sector, and filling costs	Actual-sector comparison and exact periods

The charged certificate has a genuine finite stopping interpretation: strictly separated enclosures in (8) exclude a prescribed charged coefficient. It also recovers the qualitative source implication as $\Delta, \kappa \to 0$, with $V \to a > 0$, bounded W, and $\chi_l \to 1$. The principal cost is its phase denominator. Small allowed ϵ, large charge, or large reference cost can require very small cumulant and variance

defects. The power bound in Corollary 2.3 is conservative and is not a claim of an optimal exponent in these defects.

The annular certificates expose which errors are extensive in $\log n$. They allow some scales to be less accurate and different tilts on certified heterogeneous scales, but do not grant a convergence rate merely because the local limiting theorem is qualitative. The source's full analytic-entry norm cannot be certified by finitely many real moments. Similarly, an exact finite matrix does not verify every allowed geometry, arbitrary guard data, or the continuation to b_c.

Useful further work is sharply defined. One could derive genuine finite-box bounds on Δ_m and the mixed cumulant, with all boundary comparisons included, then optimize the exact q_ϵ threshold rather than its relaxed power bound. A second task is a certified annular discretization with a full physical residual and a profile-independent admissible tilt range. A third is an actual multiscale estimate for the magnetic $\eta_{m,j}$, retaining exact support and dependence factorization. None is supplied by the finite checks below. The original source's radial coefficient and critical endpoint are not replaced by finite rank, numerical fits, or an unproved rate for the Gaussian limit.

9 Evidence and contribution assessment

All eleven cached public text files were literally read in bounded displays: 4317 lines, comprising the main file, five scientific section files, bibliography, source-reference table, two figure sources, and README. Figure code and captions were read, but the original figures were not separately rendered. Reading every line is not an independent proof audit of every imported companion lemma. In particular the companion's full localization and analytic-map proofs were not freshly read. No substantive source correction is established here.

The private original scope is 875 distinct lines in three files: the weak-operator report at lines 1–177 and 492–579; the entire 373-line outer transform chapter; and the gauge report at lines 1–96, 154–203, 454–530, and 632–645. Unread complements and endpoint hashes are recorded per file. The transform chapter's child inputs and the wider private corpus are explicitly outside the read claim. No confidential material or private titles were sent to public search.

The accompanying author-written exact program passed 2484 cases and 16387 assertions, using rational and symbolic arithmetic. Case units are named parameter tuples, actual finite laws, graphs, profiles, tilt lists, reference laws, or symbolic integral groups; an assertion is one explicit check. It checks the feasible charge envelope, root and monotonicity algebra, raw two-copy expansion, Fejer coefficients, trapezoid identities, unequal and clipped tilts, actual graph flow and gauge identities, period obstructions, independent Skellam moments, and affine residuals.

The raw replica fixtures are actual finite symmetric laws, but they do not assert conditional Gaussian parity independence. The threshold envelopes are not asserted to occur in an XY box. Independent Skellam cumulants are not substituted for coupled-height cumulants. No foreign checker was run, and independent review checks are not included in the author totals.

The assessment is C3, independently of applicability L3: the explicit defect-to-charge theorem supplies a nontrivial quantitative refinement of a specific source contradiction, with a feasible rational threshold and phase-test consequences. The annular ledgers and sharp primary prefactor make its error requirements usable. Individual Fourier, projection, flow, and gauge identities are classical supporting tools, not separately claimed breakthroughs. These local increments do not establish a new global BKT or critical-exponent theorem, and no historical priority is asserted.

References

[1] OpenAI, *The critical correlation exponent of the planar XY model*, September 24, 2026, cached manuscript at pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Scientific sections `20-heights`, `30-pins`, `40-magnetic`, `50-openness`, `60-critical`.

[2] D. van Engelenburg and M. Lis, *An elementary proof of phase transition in the planar XY model*, Commun. Math. Phys. **399** (2023), 85–104. arXiv:2110.09465v2; journal DOI.

[3] P. Lammers, *A dichotomy theory for the height functions of the BKT transition*, arXiv:2211.14365v4, July 25, 2026.

[4] T. Durand and P. Lammers, *The BKT transition and surface tension differentiability*, arXiv:2605.28473, May 27, 2026.

[5] O. Regev and N. Stephens-Davidowitz, *An inequality for Gaussians on lattices*, SIAM J. Discrete Math. **31** (2017), 749–757. arXiv:1502.04796v3.

[6] M. Subbarao, *Report 8: PDE-First Ram Formulations: Exact Green and Duhamel Equivalence, Boundary Traces, Compatible Variational Panels, and Nonlinear Algorithms*, September 18, 2026. Private original, Sections 2–3 and 10, scoped passages stated above.

[7] M. Subbarao, *RamCS Transforms and Operator Spectra*, Ram Calculus, Volume 1 revised, Chapter 7. Private original outer chapter, 373 lines; child inputs not read.

[8] M. Subbarao, *Ram Green Operators and Sparse Tree Moments for Local Heights on Semistable Curves*, TR-meta5-math, October 3, 2026. Private original, general gauge and exact pairing-under-elimination proofs and selected context.